## Experiment Name:

**Introduction to Cisco Packet Tracer and Basic PC-to-PC Connectivity**

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## Objectives:

- Learn how to use **Cisco Packet Tracer** to build and test networks.
- Understand how two computers (PCs) can connect and communicate directly.
- See how data moves between devices in a simple network.

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## Introduction:

Cisco Packet Tracer is a simulation software made by **Cisco Systems**. It helps users create and test virtual computer networks without using real hardware.A **computer network** lets devices share information and resources. The simplest type of network can be made by connecting **two PCs** directly with a cable.Cisco Packet Tracer also allows you to **watch how data packets travel** between devices in both **Real-Time** and **Simulation** modes, which helps you understand how networks actually work.

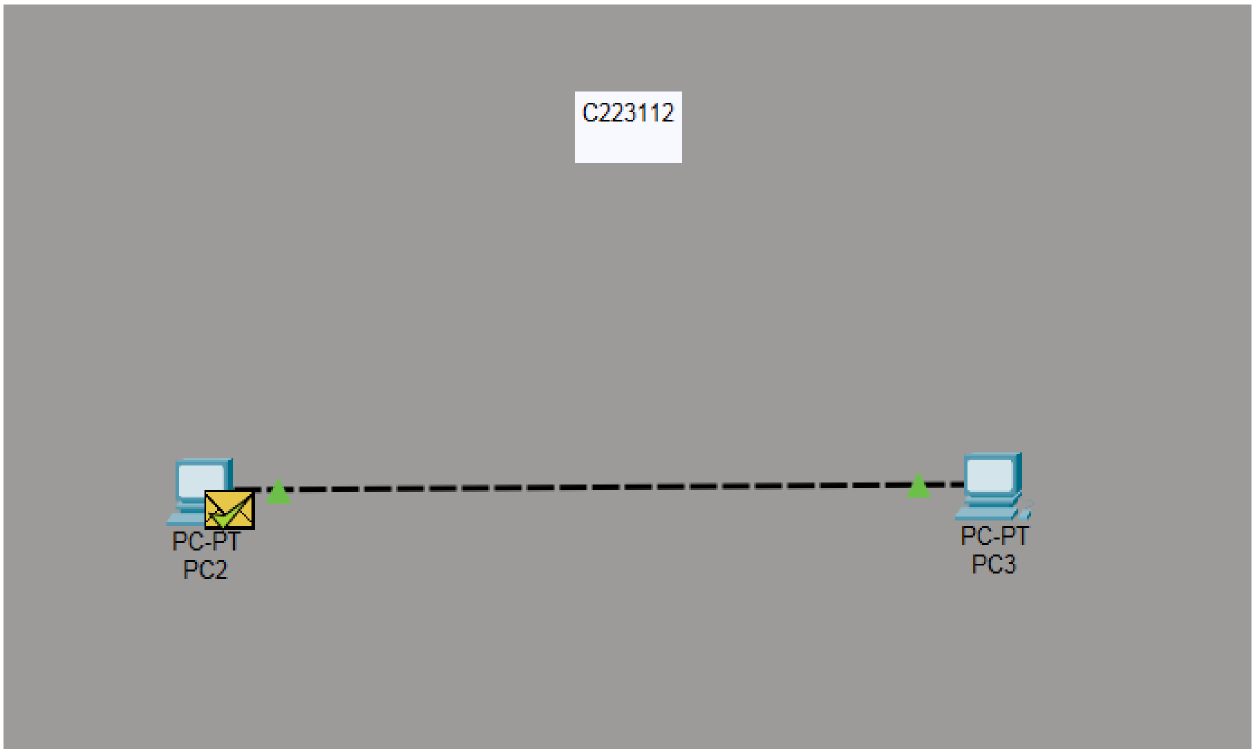
## Equipment and Tools:

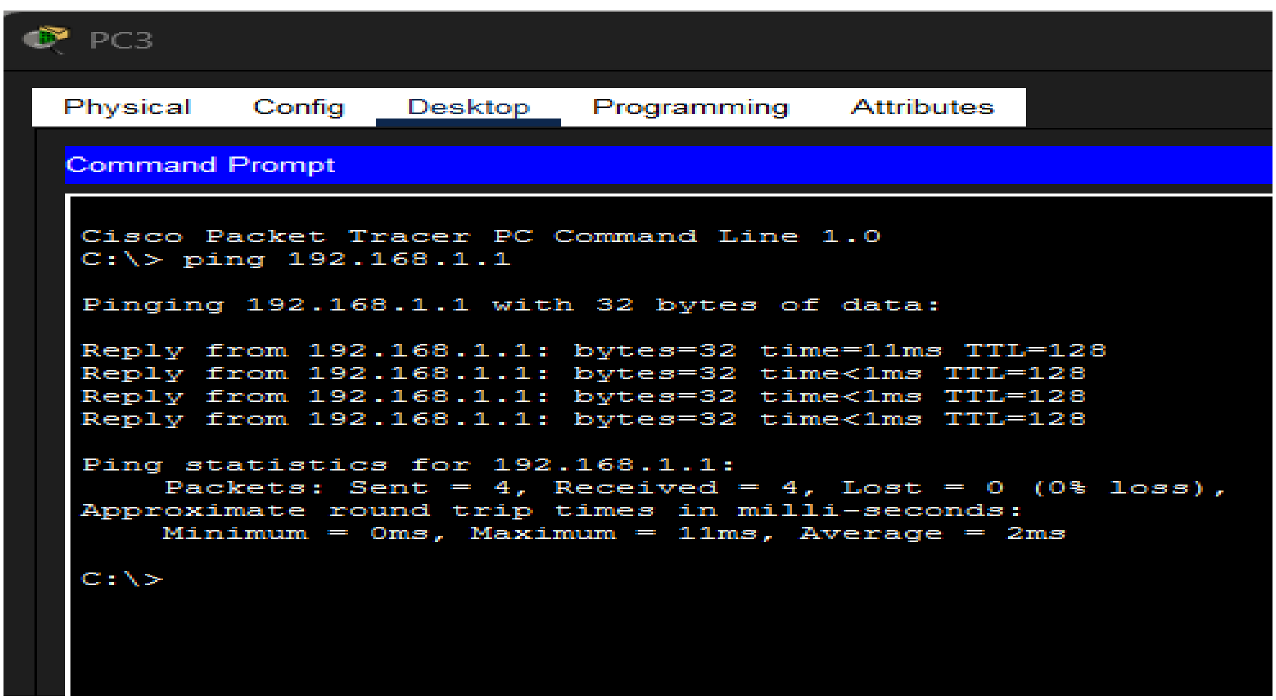
- **Software:** Cisco Packet Tracer
- **Devices:** 2 PCs
- **Cable:** Copper Cross-Over Cable
- **Commands:**
    - `ipconfig` → Check IP address configuration
    - `ping` → Test connection between two PCs

## Steps (Experimental Setup):

1. Open **Cisco Packet Tracer**.
2. Place **two PCs** on the workspace.
3. Connect both PCs with a **Copper Cross-Over Cable**.

4. Assign IP addresses:
   - PC0 → IP: 192.168.1.1
   - PC1 → IP: 192.168.1.3
5. Open the **Command Prompt** in one PC and use the `ping` command to test connectivity to the other PC.
6. Observe how packets move between PCs in **Real-Time** and **Simulation** modes.



## Result and Analysis:

Both PCs were connected successfully using a cross-over cable. After assigning the correct IP addresses, the **ping** command showed successful replies, meaning the two PCs could communicate properly. In simulation mode, the packet flow confirmed that data was sent and received between the PCs.

## Discussion:

The experiment proved that two PCs can communicate directly when they are properly connected and configured with valid IP addresses using Cisco Packet Tracer.

## Experiment Name:

**Creating a Small LAN and Comparing Hub and Switch Operation**

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## Objectives:

- To extend a simple two-PC network into a small **Local Area Network (LAN)** with multiple PCs.
- To understand and compare how a **hub** and a **switch** manage data traffic between devices.

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## Introduction:

A **Local Area Network (LAN)** connects multiple computers within a small area, such as a classroom or office, to share data and resources.
In this experiment, we build a small LAN using multiple PCs connected through a **hub** and a **switch**.

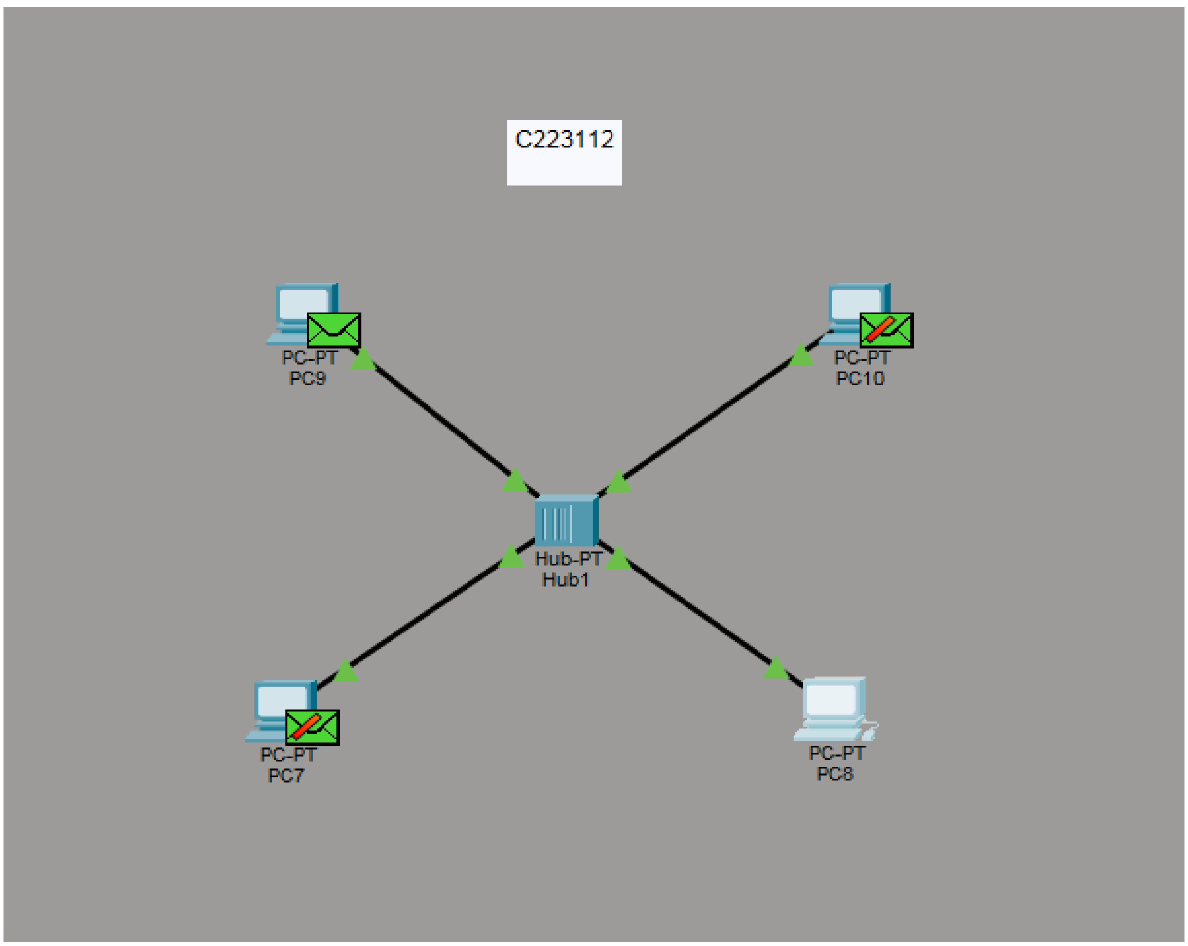
A **hub** sends data to **all connected devices**, while a **switch** sends data **only to the specific device** that needs it. This helps us see why switches are faster and more efficient in handling network traffic.

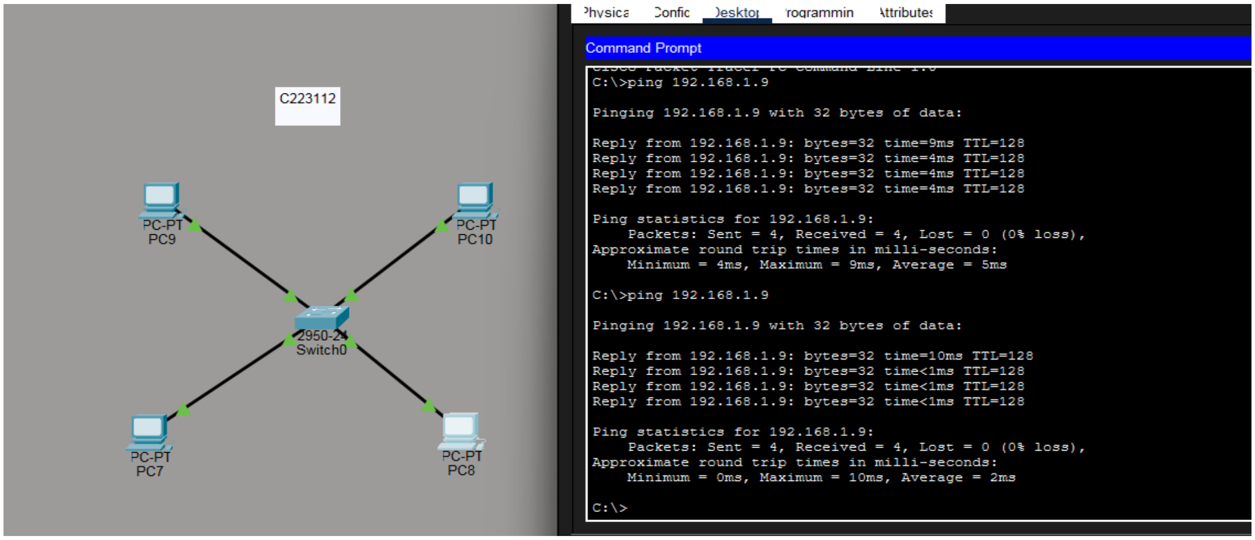
## Equipment and Tools:

- **Simulator:** Cisco Packet Tracer
- **Devices:** 3 or more PCs, 1 Hub, 1 Switch
- **Cables:** Copper Straight-Through Cables
- **Commands:**
  - `ipconfig` → to check IP configuration
  - `ping` → to test connectivity

## Steps (Experimental Setup):

1. Open **Cisco Packet Tracer**.
2. Place 3–4 PCs, one **Hub**, and one **Switch** on the workspace.
3. Connect all PCs to the **Hub** using straight-through cables.
4. Assign IP addresses to all PCs in the same network range.
5. Use the **ping** command between PCs to test communication.
6. Observe how the **Hub** broadcasts data to all PCs.
7. Replace the Hub with a **Switch** and test again.
8. Observe how the **Switch** sends data only to the destination PC.






## Result and Analysis:

All PCs were connected through a switch and successfully communicated using the **ping** command.
In **Simulation Mode**, it was observed that the **switch sent data only to the destination PC**, not to all devices. This shows that the switch learns the MAC addresses of connected devices and forwards data only where needed, making communication faster and reducing unnecessary network traffic.

## Discussion:

A **switch** is more efficient than a **hub** because it transfers data only to the intended device. This improves network speed, reduces congestion, and makes LAN communication more reliable.

Problems faced:

**Device Setup Mistakes:**
Some devices were not powered on or connected properly in Cisco Packet Tracer, which stopped the ping from working.

**Understanding Packet Flow:**
It was sometimes difficult to clearly observe how packets traveled between devices in **Simulation Mode**

## Experiment Name:

**Simulation of Network Topologies – Mesh, Ring, Star, and Bus**

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## Objectives:

- To create and understand different types of **LAN topologies**: mesh, ring, star, and bus.
- To learn how network **structure affects data flow** and **reliability**.

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## Introduction:

A **network topology** is the way computers and devices are connected in a network. It can be physical (how devices are actually connected) or logical (how data flows).
The type of topology used affects the **speed, reliability, cost, and maintenance** of the network.

## Equipment and Tools:

- **Software:** Cisco Packet Tracer
- **Devices:**
  - 4–5 PCs (End Devices)
  - Switches (Cisco 2960)
- **Cables:** Copper Straight-Through and Cross-Over (as needed)
- **Commands Used:**
  - `ipconfig` → to view and confirm IP addresses
  - `ping` → to test communication between devices

## Experimental Setup:

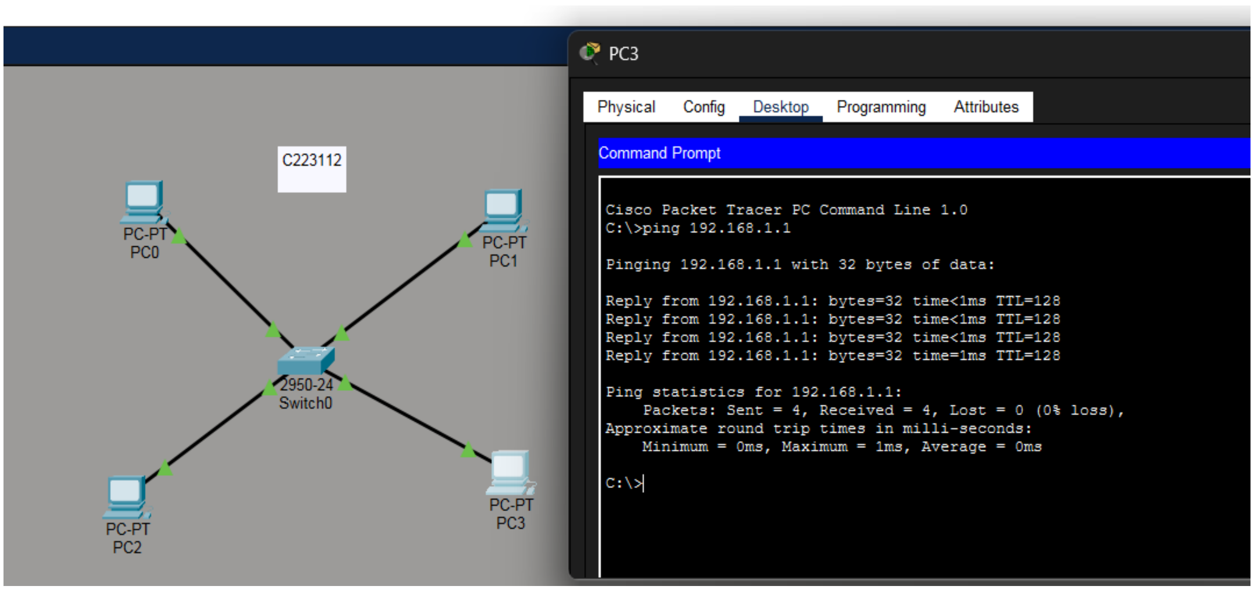
1. **Open Cisco Packet Tracer.**
2. **Place 4–5 PCs** on the workspace.
3. **Create the following topologies one by one:**
   - **Star Topology:**
     Connect all PCs to a **central switch** using **Straight-Through cables**.
   - **Mesh Topology:**
     Connect **each PC to every other PC** using **Cross-Over cables**.

- **Ring Topology:**
  Connect PCs in a **circular loop** using **Cross-Over cables**.
- **Bus Topology:**
  Connect all PCs to a **single common line (bus)** using a straight connection.

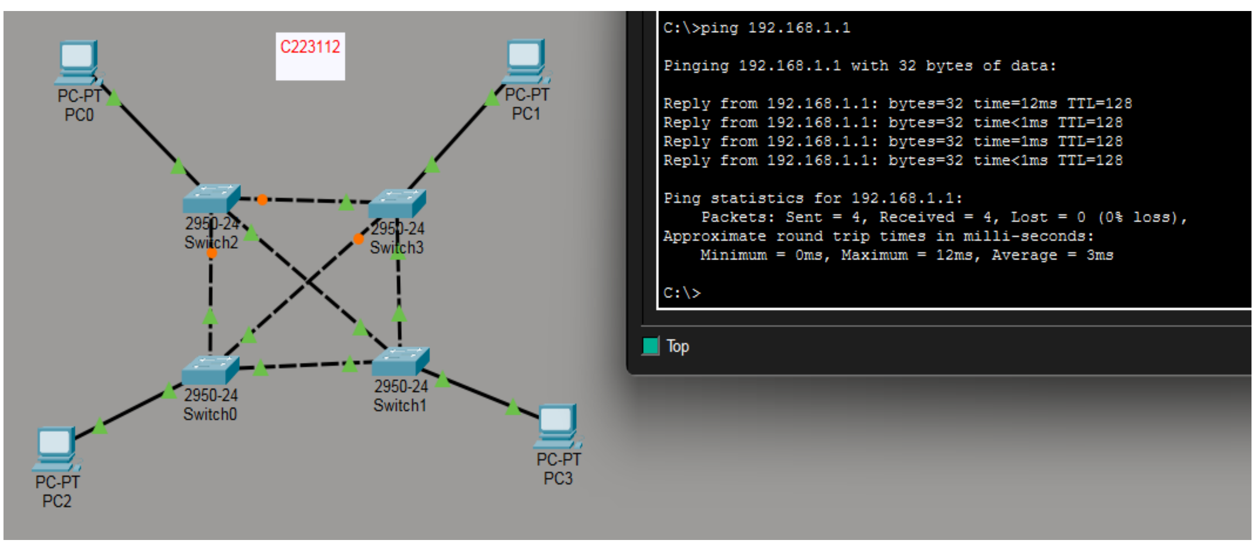
4. **Assign IP addresses** to all PCs in the same network range.

5. **Use Command Prompt on each PC** to check IP configuration using.

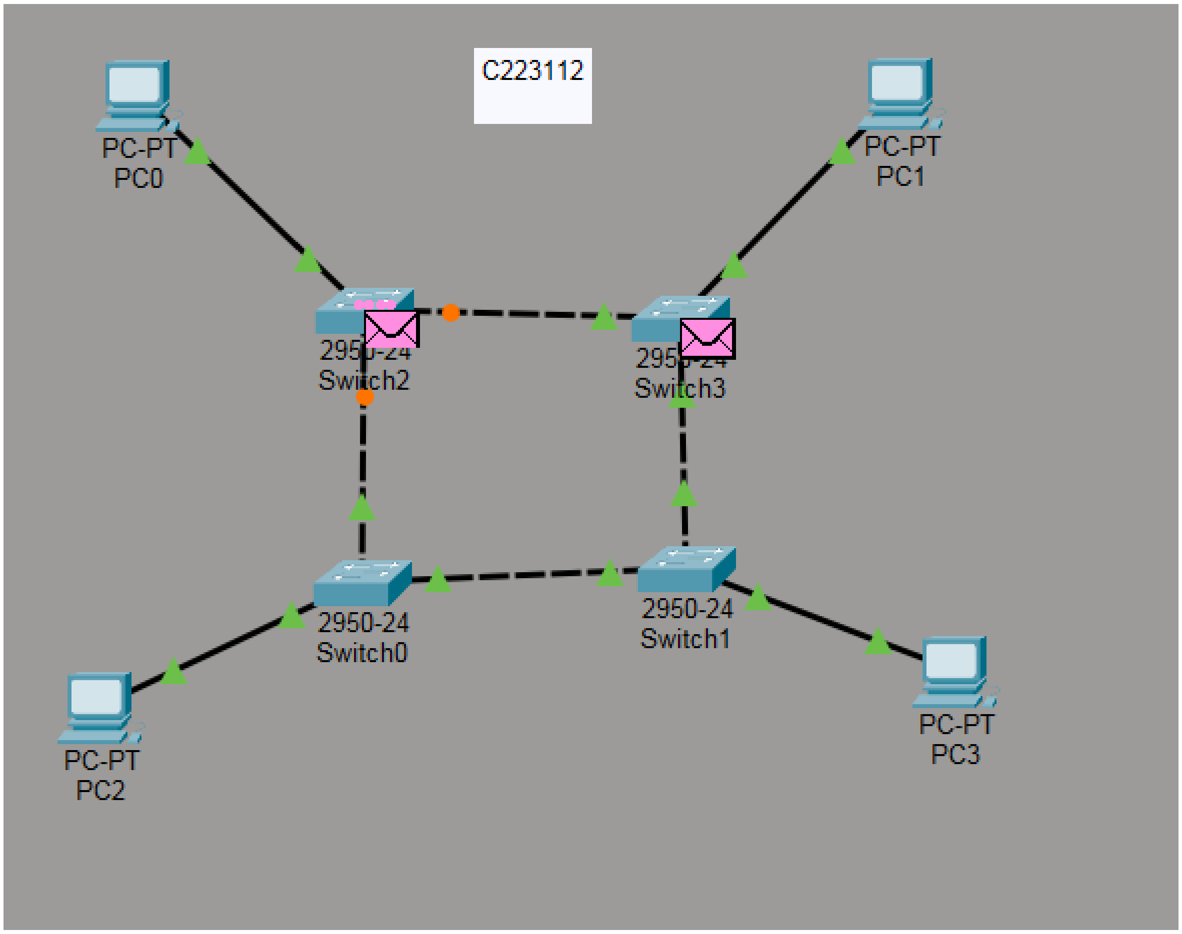
## STAR Topology



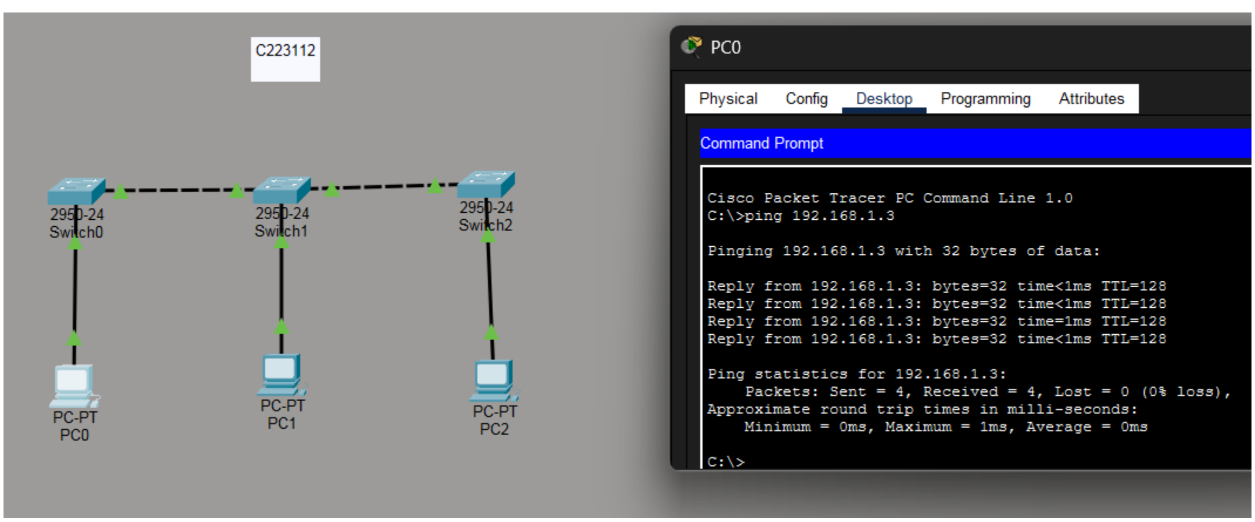

## MESH Topology



# RING Topology




# BUS Topology



## Result and Analysis:

All four topologies — **Star**, **Mesh**, **Ring**, and **Bus** — were made successfully in **Cisco Packet Tracer**. The **ping** command worked in all cases, showing that the PCs could communicate properly.

In the **Star topology**, data passed through the central switch and worked even if one PC failed.
In the **Mesh topology**, every PC was connected to all others, so it was very reliable but used many cables.
In the **Ring topology**, data moved in a circle from one PC to another; if one link broke, the network stopped working.
In the **Bus topology**, all PCs shared one main line; it was simple but caused data collisions if many PCs sent data together.

This shows that each topology has different advantages and disadvantages in data flow and reliability.

## Discussion:

In this experiment, different network topologies — **Star, Mesh, Ring, and Bus** — were created and tested in **Cisco Packet Tracer** to see how they work and how data travels between devices.

During the experiment, a few problems were faced such as:

- **Wrong IP addresses or subnet masks**, which caused ping failures.
- **Forgetting to power on devices** or connect them properly.
- Difficulty in **observing packet movement** in Simulation Mode for complex topologies like mesh and ring.

## Experiment Name:

**Network Performance Analysis using Ping Command – Throughput and Delay**

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## Objectives:

- To check **network performance** using the **ping** command.
- To measure and analyze **delay** and **throughput** in different network setups.
- To understand how **network design** affects the **speed and efficiency** of data transfer.

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## Introduction:

The **ping command** is a simple and useful tool in networking that tests whether two devices can communicate with each other. It measures how long it takes for a message (packet) to go from one device to another and back — this is called the **Round Trip Time (RTT)**.

Using ping, we can find out how much **delay** there is in the network and if any packets are lost during transmission. By checking this along with the **throughput**, we can understand how well the network performs.

• Delay (ms) : Time taken for a packet to travel from source to destination and back.

• Transmission Time (ms) : Time taken to put all bits of a packet on the wire.

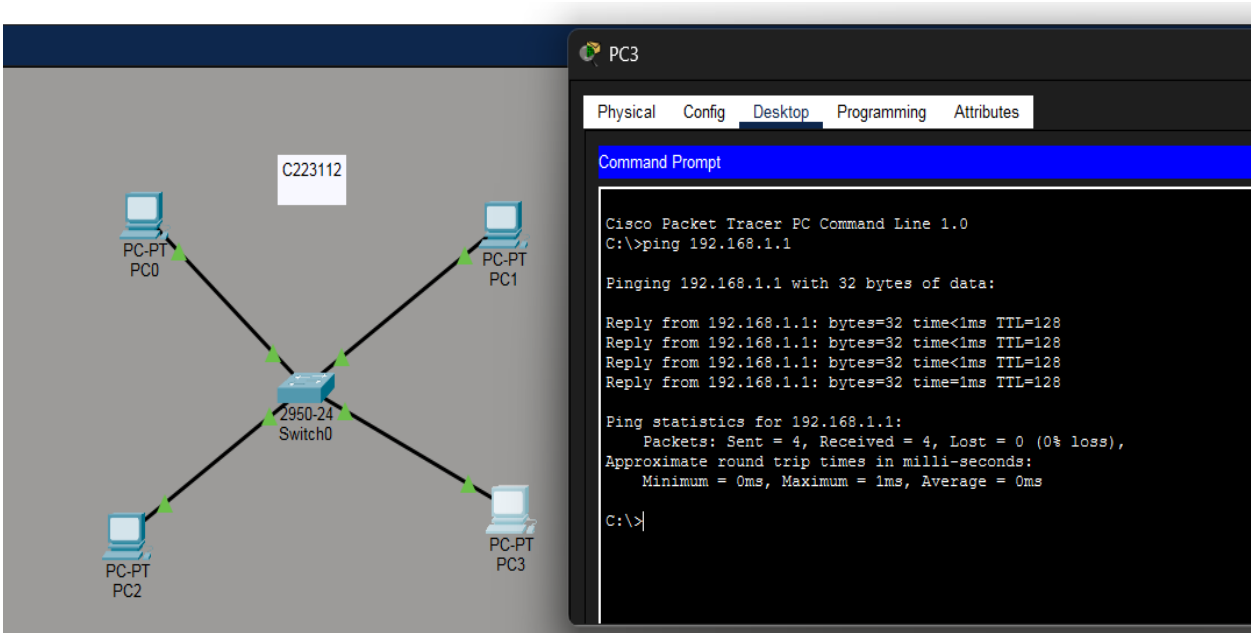
• Throughput (bps) : The actual rate of successful data transfer through the network.

Throughput = Data Size / Transmission Time

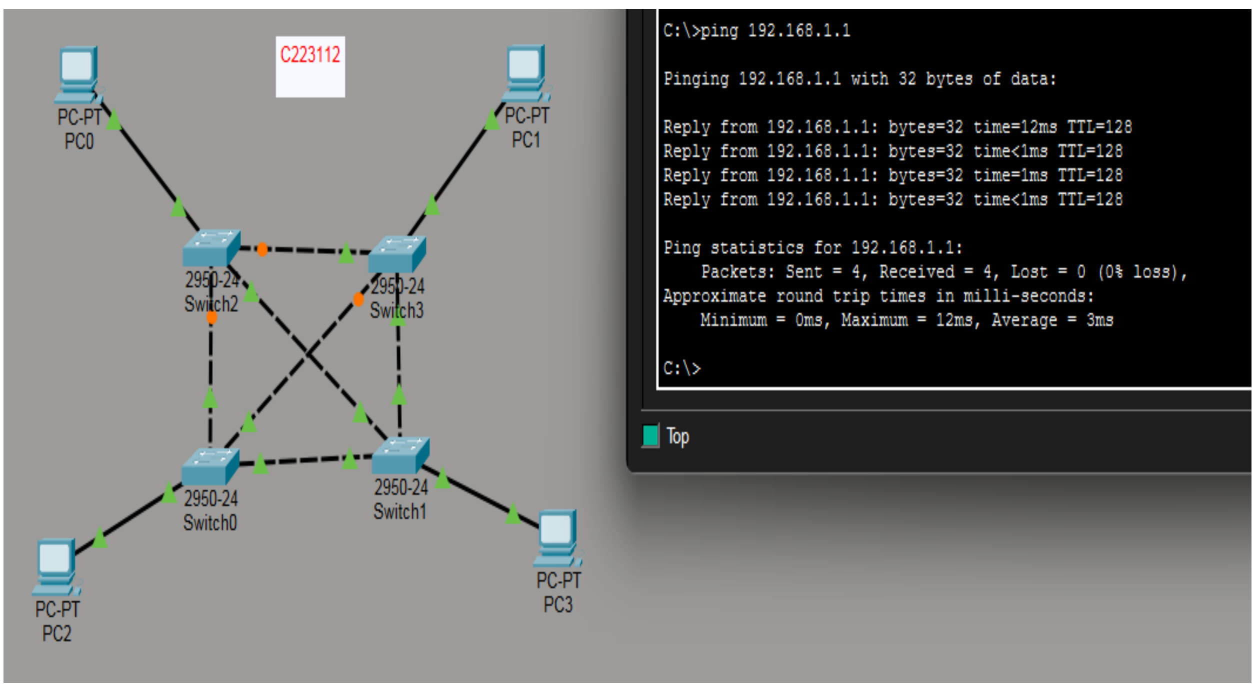
## Equipment and Tools:

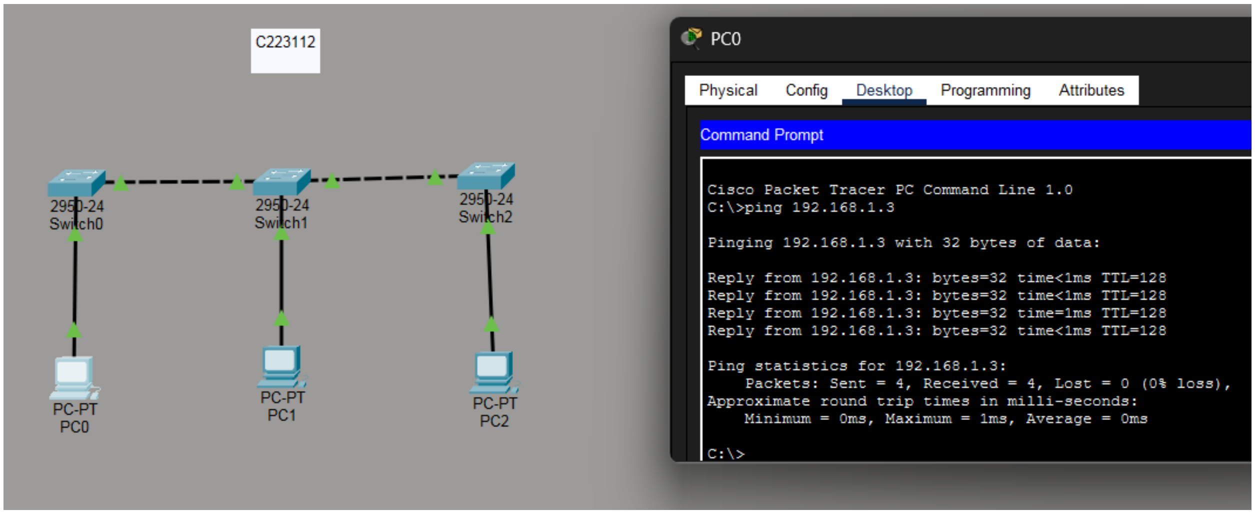
- **Software:** Cisco Packet Tracer
- **Devices:** PCs, Switches, and Routers
- **Cables:** Copper Straight-Through and Cross-Over Cables
- **Commands Used:**
  - `ping` → to check connectivity and measure delay
  - `show ip` → to verify IP configuration and routing
  - **Simulation Mode Tools** → to observe packet flow and timing

# STAR Topology




# Mesh Topology






| Network Type | Avg Delay (ms) | Throughput (kbps) | Packet Loss (%) |
|---|---|---|---|
| STAR | 0 | 256 | 0 |
| MESH | 3 | 68.26 | 0 |
| BUS | 0 | 256 | 0 |

- **In the** Star **and** Bus **topologies, delay was 0 ms and throughput was high (256 kbps).**
- **In the** Mesh **topology, delay increased to 3 ms and throughput dropped to 68.26 kbps.**
- **No packet loss occurred in any topology.**

## Result and Analysis:

- Ping between PCs and routers was successful.
  Delay was low, and throughput was good.
  When more devices were added, delay increased, and throughput decreased.

- **Discussion:**
- The network worked properly with correct IP setup.
  Ping helped check connection, delay, and speed.
  Problems faced: wrong IP settings and missing cable connections.

## *Experiment Name:*

**Introduction to Subnetting and IP Addressing – Network, Broadcast, and Usable Hosts**

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## Objectives:

- To learn how to find subnet mask, network address, broadcast address, and usable IPs.
- To understand how subnetting helps organize and manage networks better.

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## Introduction:

Subnetting means dividing one large network into smaller parts called subnets.
It helps make the network faster, more secure, and easier to manage.

**Important terms:**

- **IP Address:** A unique number given to every device on a network.
- **Subnet Mask:** Tells which part of the IP is for the network and which is for the host.
- **Network Address:** The first IP in a subnet, used to identify that subnet.
- **Broadcast Address:** The last IP in a subnet, used to send data to all devices in that subnet.
- **Usable IPs:** All the IPs between the network and broadcast address that can be given to devices.

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## Equipment and Tools:

- **Simulator:** Not required
- **Tools:** Calculator & worksheet for subnetting calculations

## QUESTION:

1) Suppose an ip address of a host is 192.168.5.85/27

1. What is the subnet mask 2. Block size 3. Number of subnet 4. Number of host 5. What are the subnet addresses 6. First and last valid host 7. Broadcast address.

## Solution:

### *1) For IP 192.168.5.85/27*

1. **Subnet mask:** /27 = 255.255.255.224
   (224 = 11100000 )
2. **Block size :** 256 − 224 = 32 → **32**
3. **Number of subnets (within a /24):** borrowed bits = 3 → 2^3 = 8 subnets
4. **Number of hosts per subnet:** host bits = 5 -> 2^5 − 2 = 30 usable hosts
5. **Subnet addresses :**
   192.168.5.0, 192.168.5.32, 192.168.5.64, 192.168.5.96, 192.168.5.128, 192.168.5.160, 192.168.5.192, 192.168.5.224
6. **First and last valid host in that subnet:**
   - First usable: **192.168.5.65**
   - Last usable: **192.168.5.94**
7. **Broadcast address (for that subnet): 192.168.5.95**
8. **Network address:** 192.168.5.64

Q2) 192.168.5.85/20 is an IP address of a host in a network.

Find the following: I. Subnet mask II. Network Address III. Broadcast address IV. First usable address V. Last usable address

### SOLUTION:

- **Subnet Mask: 255.255.240.0**
- **Network Address: 192.168.0.0**
- **Broadcast: 192.168.15.255**
- **First Usable: 192.168.0.1**
- **Last Usable: 192.168.15.254**

## Result and Analysis:

Subnetting was done successfully using given IPs.
We found the subnet mask, network address, broadcast address, and usable host range correctly.
The results showed how IPs are divided into smaller networks for better management and less waste.

## Discussion:

This experiment helped understand how subnetting organizes a large network into smaller parts.
It also showed the difference between network, broadcast, and usable IPs.